by hard minerals and get fossilized, which are called casts. Hardened faecal matter termed as coprolites occur as tiny pellets. Analysis of the coprolites enables us to understand the nature of diet the pre-historic animals thrived on.

Visit any museum nearer to your school with your teacher and identify the bones of different animals including mammals. The famous Egmore Museum is in Chennai.

### 6.4.2 Evidences from comparative anatomy

Similarities in structure between groups of organisms are accepted as indicators of relationship. For example, a comparative study of the forelimbs of different vertebrates exhibits a fundamental plan of similarity in structure. These relationships can be studied under homologous organs, analogous organs, vestigial organs, connecting links and atavistic organs.

### Homologous structures

In vertebrates, comparative anatomical studies reveal a basic plan in various structures such as fore limbs and hind limbs. Fore limbs of vertebrates exhibit anatomical similarity with each other and is made of similar bones such as humerus, radius, ulna, carpals, metacarpals and phalanges.

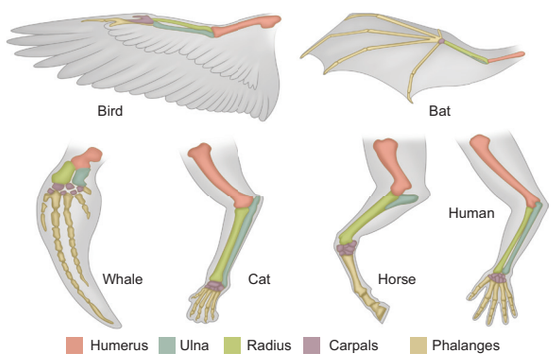



**Fig. 6.2 Forelimbs of terrestrial vertebrates to show homology**

Structures which are similar in origin but perform different functions are called homologous structures that brings about **divergent evolution (Fig 6.2).**

Similarly the thorn of *Bougainvillea* and the tendrils of *Curcurbita* and *Pisum sativum* represent homology. The thorn in former is used as a defence mechanism from grazing animals and the tendrils of latter is used as a support for climbing.

### Analogous structures

Organs having different structural patterns but similar function are termed as analogous structures. For example, the wings of birds and insects are different structurally but perform the same function of flight that brings about **convergent evolution (Fig. 6.3).**

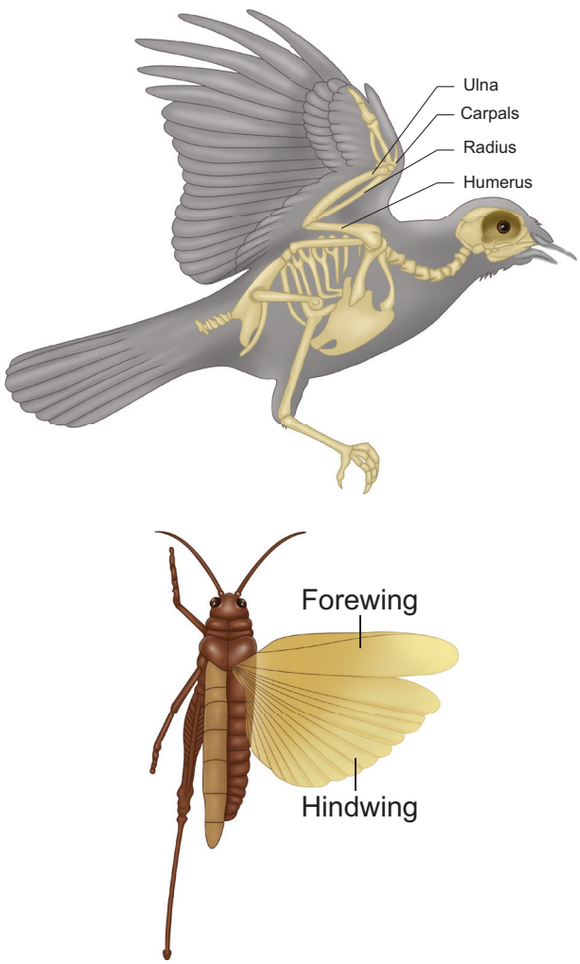



**Fig. 6.3 Comparison of insect and bird wing to show their analogy**

Other examples of analogous organs include the eyes of the Octopus and of



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